

Personal Assistant / NutriStart Foundation

Technical Case Study

Project Overview

This project started as an experiment to understand modern AI application architecture and evolved into a production-style personal assistant platform. The solution supports both local and cloud AI providers through a provider-based architecture, enabling future expansion into the NutriStart ecosystem.

Objectives

- Build a cross-platform AI assistant using Flutter.
- Support local and cloud-hosted AI models.
- Create a provider abstraction layer for extensibility.
- Deploy a public web version using GitHub Pages and Render.
- Establish a foundation for future Android and NutriStart development.

Architecture

Flutter (Desktop/Web) → FastAPI Backend → Provider Factory → AI Providers (Ollama, Gemini, OpenAI-ready).

Technology Stack

Frontend: Flutter, Material UI, Shared Preferences

Backend: Python, FastAPI, Uvicorn

AI Providers: Ollama, Qwen2.5-Coder, Gemini

Deployment: Render, GitHub Pages

Configuration: Environment Variables, Provider Settings

Key Engineering Decisions

1. Provider Pattern: Decoupled the application from any specific AI model.
2. Configuration Driven Design: Provider and backend URL managed through settings.
3. API-Centric Architecture: Flutter communicates only with FastAPI.
4. Cloud-Ready Deployment: Designed for migration from local execution to hosted environments.

Challenges & Resolutions

Windows Desktop Build Issues

Resolved Visual Studio, CMake, Flutter desktop tooling, and dependency compatibility issues.

Provider Switching

Implemented a provider factory enabling seamless switching between Ollama and Gemini.

Cloud Deployment

Migrated backend to Render while keeping API keys secure using environment variables.

GitHub Pages Deployment

Resolved Flutter web routing issues through correct base-href configuration.

Key Learnings

- Understanding the distinction between model providers and models.
- Building maintainable AI systems through abstraction layers.
- Managing environment-specific configurations.
- Deploying production-style applications using modern cloud tooling.
- Handling CORS, API security, and provider availability concerns.
- Structuring applications for future scalability and mobile deployment.

Future Roadmap

- Android Application Release
- Chat History & Conversation Persistence
- User Authentication
- Additional Providers (OpenAI, DeepSeek, Groq)
- NutriStart Integration
- Cloud Database & Memory Layer